

April 4, 2025

**Statement of Work in Relation to Renewing
the Maintenance Agreement for the FINE/Turbo Software**

As part of the planned and ongoing research activities under the NASA Advanced Air Transport Technology (AATT) Project and the Commercial Supersonic Technology (CST) Project, the FINE/Turbo software is used for steady and unsteady aerodynamic simulations that are necessary for fan, open fan, open rotor, and turbomachinery noise prediction. The simulation results are compared with wind tunnel and rig data when possible. Ultimately, in conjunction with NASA fan and open rotor noise prediction codes, these simulations are used to (1) assess acoustic performance of candidate fan and open rotor designs, and (2) identify potential shortcomings in prediction tools and devise model/code improvements that may be needed.

LTVA Branch, NASA Glenn